

Pre-registration v1

The Causal Safety Audit: Counterfactual Stress-Testing for Deep-Learning MRI Reconstruction

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Status at deposit. No reconstruction has been run. No pilot has been run. No experimental result of any kind exists at the time of this deposit. The thresholds in Section 4 were fixed from convention and prior literature, with no contact with reconstruction output. The code as of this deposit is archived at Zenodo concept DOI 10.5281/zenodo.22648224 (release v0.1.0, version DOI 10.5281/zenodo.22648225).

1 Study rationale and question

Deep-learning MRI reconstruction is evaluated almost universally with global image scores (PSNR, SSIM) and with data-consistency residuals. Both are insensitive, by construction, to content the measurement did not constrain: a reconstruction may remove a small lesion and still report an excellent global score and a residual at the noise floor. This study asks whether that failure mode occurs at rates of clinical relevance, and whether the standard metrics detect it.

The audit inserts a lesion on the *measurement* side of the true multi-coil forward operator, $Y_{\text{cf}} = Y + \mathbf{A}\ell$, so that the factual and counterfactual arms share the measured noise realisation, sampling mask and coil sensitivities exactly. Both arms are reconstructed with the same model and the same seed. Every lesion-attributable quantity is a within-pair difference.

2 Hypotheses

- H1** The silent erasure rate for lesions of $\leq 10 \text{ mm}^3$ at $R \geq 6$ exceeds the pre-specified rate of clinical relevance π_0 in at least one model family. *Directional; one-sided.*
- H2** Global PSNR and SSIM do not separate erased from preserved pairs: the standardised mean difference in each score lies within the equivalence margin, *and* the AUC for classifying erasure from the score alone lies within the margin about 0.5. *Equivalence hypothesis; both conditions required.*
- H3** ROI contrast and per-lesion detectability z do separate erased from preserved pairs, with classification AUC above the pre-specified floor. *Directional.*
- H4** The false-structure count in the lesion-absent arm increases with acceleration R , and is highest for the generative family. *Directional; trend and family contrast.*

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H2 is a null result by construction. It is therefore tested as an equivalence hypothesis against margins fixed in advance. A merely non-significant difference will be reported as inconclusive and *not* as support for H2.

3 Design

Data. The multi-coil brain subset of fastMRI, which provides fully sampled raw k -space with coil sensitivities recoverable from the calibration region. Sequences are T1-weighted, T2-weighted and FLAIR axial slices at 1.5 T and 3 T. Retrospective undersampling of already-processed magnitude images is excluded as a validity requirement. Lesion annotations come from fastMRI+; slices used for insertion are those with no annotation within the audited region.

Synthetic lesion bank. Spherical or ellipsoidal perturbations with volumes 1, 5, 10 and 27 mm³; three contrast levels spanning the fastMRI+ interquartile range for the matching sequence; both signs of contrast where the sequence admits them (hyperintense on FLAIR and T2, hypointense on T1). Sites are drawn from white matter, deep grey matter and the cortical ribbon, avoiding ventricles and skull.

Models under audit. Three families, audited *as released*: (i) End-to-End Variational Network (physics-based unrolled, learned data consistency); (ii) an image-domain complex U-Net baseline; (iii) a score-based diffusion reconstructor with a data-consistency step, evaluated with common random numbers across the two arms of each pair. No model is retrained on the lesion bank, and no lesion-bearing slice enters any training or calibration step.

Conditions. $R \in \{4, 6, 8\}$ with equispaced Cartesian masks; random masks as a secondary condition.

Unit of analysis. The lesion pair. Pairs are nested in slices and slices in subjects, and one host slice carries several lesions, so pairs are not independent.

4 Pre-specified thresholds

These values are fixed by this deposit. They are normative choices drawn from convention and prior literature, not estimates from data. They are recorded machine-readably in `configs/thresholds.yaml` in the archived repository.

Symbol	Value	Meaning and justification
z_{det}	3.0	Reference detectability required for a lesion to enter the erasure analysis. The channelised Hotelling observer per-lesion z is a normal deviate; 3.0 is the conventional “clearly visible” anchor. The Rose criterion (5.0) was considered and rejected: it would exclude most of the $\leq 10 \text{ mm}^3$ bank that H1 concerns, and so would answer a different question.
z_{miss}	2.0	Reconstruction detectability below which the lesion counts as missed. Set strictly below z_{det} by design: pairs falling in the indeterminate band $[2.0, 3.0)$ are counted as <i>neither</i> erased nor preserved, so borderline pairs cannot inflate the erasure rate.
z_{fs}	4.0	False-structure threshold, in standard deviations of the null-space error over background tissue. Candidates are screened over many connected components per slice; 4sd controls the per-image false-positive count. A candidate must additionally have a volume of at least the smallest lesion in the bank.
π_0	0.05	Silent erasure rate taken as the floor of clinical relevance for sub- 10 mm^3 lesions at $R \geq 6$. A domain judgement, not an estimate.

Definition of erasure. A pair is *erased* when

$$z_i^{\text{ref}} \geq z_{\text{det}} \quad \text{and} \quad z_i^{\text{recon}} < z_{\text{miss}}. \quad (1)$$

A pair is *silently erased* when, in addition, the global PSNR of the lesion-present reconstruction lies within the central 95% interval of PSNR over lesion-absent reconstructions of the same slice set under the same model and R , so that the reported score gives no indication that anything is missing.

Other fixed analysis constants. Equivalence margin on the standardised mean difference, ± 0.2 ; AUC equivalence interval for H2, $[0.45, 0.55]$; AUC floor for H3, 0.80; family-wise $\alpha = 0.05$ with Holm multiplicity correction; one-sided $\alpha = 0.025$ for H1; PSNR interval level 0.95; regularisation rule $\lambda = \sigma_\eta^2 / \text{var}(|X|)$ with a sensitivity analysis across one decade either side; target half-width of the 95% interval for a rate of π_0 , 0.02.

5 Statistical analysis plan

Primary endpoint. The silent erasure rate for lesions of $\leq 10 \text{ mm}^3$ at $R \geq 6$, per model.

Dependence. Every rate is estimated with a confidence interval from a cluster bootstrap that resamples *subjects*. Every model-level contrast is fitted as a mixed-effects model with random intercepts for subject and slice. The paired structure of the two arms is preserved throughout.

H1. Tested against $\pi_0 = 0.05$ with a one-sided cluster-bootstrap test at $\alpha = 0.025$, model by model.

H2. For each of PSNR and SSIM, the standardised mean difference between erased and preserved pairs is tested with two one-sided tests against a margin of ± 0.2 , using cluster-robust standard errors; equivalence is declared only if both tests reject. The AUC of the global score as a classifier of erasure is estimated with a subject-level bootstrap interval and equivalence to 0.5 is declared if that interval lies within $[0.45, 0.55]$. Both conditions are required.

H3. Compares the AUC of z , of ROI contrast and of t_N as classifiers of erasure against the floor of 0.80 and against the AUC of PSNR, with subject-level bootstrap intervals for the paired difference.

H4. Fits the false-structure count as a mixed-effects Poisson model with R as an ordered factor and family as a fixed effect, and tests the trend in R and the family contrast.

Multiplicity. H1 to H4 form one family of four primary tests, controlled with the Holm procedure at a family-wise level of 0.05. Analyses by lesion volume, contrast, site and sequence, the sensitivity analyses over λ and observer internal noise, and the Case B analysis on real annotated lesions are *secondary* and are reported with intervals, not tests.

Reporting. All per-pair readouts are released as a table carrying subject, slice, site, volume, contrast, model, R , mask and seed identifiers, together with provenance columns (git commit, config hash, checkpoint hash, data checksums), so that every figure and every rate in the paper can be recomputed from it.

6 Sample size and the design pilot

The lesion bank is sized so that the half-width of the 95% interval for a rate of π_0 is at most 0.02 after inflation by the design effect $1 + (m - 1)\rho$, with m the pairs per subject and ρ the intra-subject correlation.

ρ is not known at the time of this deposit and will be estimated from a design pilot of 20 subjects. The following constraints are fixed *now*:

1. The sizing *rule* above is fixed by this deposit. Only the realised number of pairs per model-by- R cell remains to be computed from it.
2. The pilot estimates ρ **and nothing else**. No endpoint, no erasure rate, no hypothesis test and no AUC will be computed on pilot data.
3. The 20 pilot subjects are **excluded** from the confirmatory analysis set.
4. The realised pairs-per-cell will be recorded in a dated addendum to this registration, posted *before* any confirmatory reconstruction is run.

This sequencing is deliberate. It places the timestamp on the thresholds and the analysis plan before any contact whatsoever with reconstruction output, rather than after a pilot has been seen.

7 Exclusions, stopping and deviations

Slice-level exclusions. Slices carrying a fastMRI+ annotation within the audited region are excluded from insertion. Slices failing the integrity check (calibration region unusable, coil sensitivity estimation non-convergent) are excluded and counted.

Pair-level exclusions. Pairs whose reference detectability falls below z_{det} do not enter the erasure analysis: the lesion was not detectable even in the fully sampled reference, so its absence in the reconstruction is not attributable to the model. These are counted and reported.

Indeterminate pairs. Pairs with z_i^{recon} in $[z_{\text{miss}}, z_{\text{det}})$ are neither erased nor preserved. Their count is reported per cell.

No optional stopping. The number of pairs per cell is fixed before the confirmatory run and analysis is performed once, after the full run completes.

Deviations. Every departure from this document is recorded in `CHANGELOG.md` in the archived repository, with its date and reason, and reported in the paper.

8 Outcome-neutral conditions

The following must hold for the study to be interpretable, independently of which hypotheses are supported:

1. The forward operator passes its adjoint test.
2. Lesion insertion passes the two-path equality test (insertion through the operator equals insertion in image space followed by the operator).
3. The filtered projectors satisfy the closed-form and transfer-identity tests, and $t_{\mathcal{R}} = 1$ under hard data consistency.
4. The channelised Hotelling observer does not exceed the ideal-observer bound.
5. Both arms of every pair share the seed; seeds are varied across replicates.

Failure of any of these is a defect to be fixed before the confirmatory run, not a result.

Materials

Code: <https://github.com/dmai287/csa-audit>, archived at Zenodo concept DOI 10.5281/zenodo.22648224. Data: fastMRI and fastMRI+, under their respective data-use terms. Derived lesion banks are released only if those terms permit; otherwise the generation script alone is released.